

Retrofitting a Hidden Reservoir in Fountain Birdbath

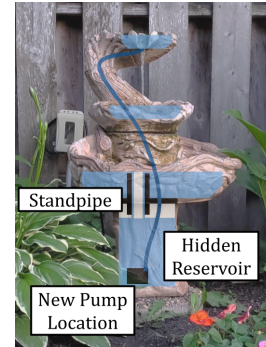
Designer / Jun 2025

Overview:

Planned a retrofit to add a hidden reservoir to a fountain-style birdbath, increasing water capacity and improving function without changing its appearance. The design included features for debris settling, extended refill intervals, and pump protection, all concealed within the existing structure. However, the fountain's fragility from freeze-thaw damage ultimately prevented safe implementation of the modification.

Tools & Skills:

Design Planning, Conceptual Sketching, Product Modification, Problem Identification and Risk Analysis



Sketch of reservoir overlaid on fountain.

Objective:

Design a retrofit to integrate a hidden reservoir into an existing fountain-style birdbath, improving water capacity without altering its appearance.

Design Highlights:

- Added buffer volume to enable debris settling and improve water quality.
- Increased water capacity to extend intervals between refills.
- Elevated head level to ensure continuous pump submersion and prevent damage.
- Concealed the reservoir within an existing fountain cavity for a seamless appearance.
- Incorporated a standpipe drain to regulate and maintain consistent water level in the primary basin.

Reflections:

Although the modification was intended to be quick and straightforward using readily available parts, the fountain's fragility from repeated freeze-thaw cycles presented a significant risk. Water trapped during a previous freeze caused the basin to shatter; despite repairs, the structure remained too fragile to safely continue with the planned modification.